Subham Sahu

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Projects

Krishi Mitra: An AI-Powered Smart Agriculture Application with Disease Prediction and Conversational Advisory System (Github)

- Developed a farmer-centric intelligent decision support system integrating real-time weather and market data to provide data-driven crop guidance and adaptive advisories.
- Architected the platform to support integration of **deep learning-based computer vision models** for automated crop disease detection and plant health monitoring.
- Explored NLP-based multilingual conversational agents to assist farmers in regional languages, enabling accessible and context-aware AI interaction.
- Designed the system modularly to incorporate Generative AI models for dynamic agronomic insight generation and automated recommendation synthesis based on environmental inputs.
- Implemented a scalable MVVM-based architecture with cloud-backed data storage, enabling future development of personalized recommendation systems using user behavior and historical trend analysis.

Stateful LLM Agent with Graph-Based Orchestration and Persistent Memory (Github)

- Designed and implemented a **stateful LLM-based conversational agent** using LangGraph and LangChain, modeling dialogue management as a Directed Acyclic Graph (DAG) for structured reasoning and controlled conversation flow.
- Engineered a graph-based state machine architecture where nodes encapsulate LLM reasoning steps and edges define transition logic, enabling modular, interpretable, and resumable agent workflows.
- Integrated SQLite as a persistence layer to implement long-term conversational memory, supporting checkpointing and recovery via thread-specific state tracking.
- Deployed ultra-fast inference using Groq LLM APIs, enabling low-latency streaming responses suitable for real-time AI applications.
- Architected the system to serve as a foundation for advanced AI agents, including multi-step reasoning workflows, autonomous task execution, and memory-augmented generative systems.

Education

- **Utkal Mani High School** — Madhabandha, Ganjam, Odisha
  Matriculation - 81% — *2019 – 2021*
- **Chikiti HS School** — Chikiti, Odisha
  Intermediate - Science (PCM) - 80.67 % — *2021 - 2023*
- **NIST University, Berhampur** — Berhampur, Odisha, India
  B-Tech - Computer Science and Technology - 9.24 CGPA — *2023 - 2027*

Skills Summary

- **Language:** Python, C++, Java, Kotlin
- **Machine Learning & Deep Learning:** Pytorch, Scikit-learn, Supervised Learning, Model Evaluation, Neural Networks
- **NLP & Generative AI:** NLTK,Langchain, LangGraph, LLM Orchestration, Memory-Augmented AI Systems
- **Data Analysis and Visualization:** Pandas, Matplotlib, Seaborn, data Preprocessing, Exploratory Data Analysis(EDA)
- **Databases:** MySQL, SQLite, Postgress
- **Tools:** Jupyter Notebook, VS Code, Git, Github

Certifications

- **Data Science and ML,DL,NLP** - Udemy
- **Competitive Programming** - Code Tantra
- **Python essential for Data Science** - Scaler
- **Data Science and Analysis** - *Summer Course at NIST University under a Data Scientist*

Club / Community Work

- Active member and event organizer at the **Data Science Club, NIST University**
- Organized and conducted **two coding competitions** in collaboration with **CodeChef**
- Assisted in planning and executing **workshops and technical events**, coordinating with teams and participants

- Developed strong **leadership, teamwork, communication, and event management** skills through these activities.

Achievements

- Awarded the **Tuition Fee Waiver (TFW)** by **AICTE** at the national level for academic excellence
- **Winner – Code Crusade**, National Tech Fest **Sankalp**, NIST University
- **Finalist – Oddo Hackathon (Offline Round)**, selected among multiple teams for presenting a problem-solving solution and competing in an on-site round

Summary

- Motivated Computer Science undergraduate with a strong interest in Artificial Intelligence, Machine Learning, and Generative AI systems. Experienced in Python-based ML development, deep learning using PyTorch, and building intelligent systems involving NLP and memory-augmented LLM architectures. Passionate about designing scalable AI solutions, experimenting with data-driven models, and applying theoretical concepts such as data structures and algorithms to real-world problems. Actively engaged in technical projects and research-oriented development, seeking a research internship at IITs to contribute to impactful AI innovations while strengthening my foundations in advanced machine learning and computational research.